

Skills Summary

Corresponding Parts of Congruent Triangles

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding a matching side

Rule. A congruence statement is read position by position. In $\triangle ABC \cong \triangle DEF$ the vertices pair as $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, so the sides pair as $\overline{AB} \leftrightarrow \overline{DE}$, $\overline{BC} \leftrightarrow \overline{EF}$, $\overline{CA} \leftrightarrow \overline{FD}$.

Matching sides are congruent, so their two expressions are *equal*: set them equal and solve.

Example. $\triangle ABC \cong \triangle DEF$ with $AB = (3x - 5)$ cm and $DE = 16$ cm. Find x .

Step 1. Want x ; the statement pairs $\overline{AB}$ with $\overline{DE}$, so the two lengths are equal — that turns the geometry into one linear equation.

Step 2. $3x - 5 = 16 \Rightarrow 3x = 21$

$$x = 7$$

Try it. $\triangle PQR \cong \triangle STU$ with $PQ = (4x + 2)$ mm and $ST = 26$ mm. Find x .

$$9 = x \text{ :check}$$

(!) Watch out: pair by *position*, not by alphabet. In $\triangle ABC \cong \triangle FED$, side $\overline{AB}$ matches $\overline{FE}$ — not $\overline{DE}$.

Skill 2 — Finding a matching angle

Rule. Matching vertices carry matching angles. In $\triangle ABC \cong \triangle XYZ$, $\angle A \cong \angle X$, $\angle B \cong \angle Y$, $\angle C \cong \angle Z$. Equal angles means equal measures, so an expression for one may be set equal to the measure of the other and solved the same way.

Example. $\triangle ABC \cong \triangle XYZ$ with $m\angle B = (5x + 8)^\circ$ and $m\angle Y = 53^\circ$. Find x .

Step 1. Want x ; B sits in the second position of both names, so $\angle B$ and $\angle Y$ correspond and their measures are equal.

Step 2. $5x + 8 = 53 \Rightarrow 5x = 45$

$$x = 9$$

Try it. $\triangle ABC \cong \triangle XYZ$ with $m\angle A = (7x - 3)^\circ$ and $m\angle X = 46^\circ$. Find x .

$$7 = x \text{ :check}$$

Skill 3 — Using CPCTC for a missing part

Rule (CPCTC). Once two triangles are known to be congruent, *every* pair of corresponding parts is congruent — including parts no one measured. Combine that with the angle sum of a triangle,

$$m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ,$$

to reach a missing measure in either triangle.

Example. $\triangle GHJ \cong \triangle KLM$ with $m\angle K = 42^\circ$ and $m\angle L = 61^\circ$. Find $m\angle J$, in degrees.

Step 1. Want $m\angle J$; J corresponds to M , so by CPCTC $m\angle J = m\angle M$ — and $\angle M$ is the third angle of a triangle whose other two measures are known.

Step 2. $m\angle M = 180 - 42 - 61$

$$m\angle J = \mathbf{77} \text{ degrees}$$

Try it. $\triangle GHJ \cong \triangle KLM$ with $m\angle K = 35^\circ$ and $m\angle L = 80^\circ$. Find $m\angle J$, in degrees.

check: $m\angle J = 65$ degrees

(!) Watch out: CPCTC is a conclusion, never a reason to start. First establish the congruence (SSS, SAS, ASA, AAS or HL); only then may you claim a pair of parts is congruent.